
Finite- K Posterior Learning with One-Draw Primal Updates

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Abstract

1 Multi-sample reasoning improves reliability by generating several candidate solu-
2 tions, increasing the chance that at least one succeeds within a limited inference
3 budget. The candidate pool’s success depends on which solver variants it con-
4 tains, so we model their distribution as a regularized posterior over solver param-
5 eters. Each posterior draw instantiates one candidate solver, and we optimize the
6 posterior to increase the chance that at least one candidate succeeds. Direct op-
7 timization of this objective requires several sampled solvers in every update, so
8 we use a Fenchel reformulation and a Bregman update to learn from one posterior
9 draw and a running summary of each problem’s failures. This keeps the number of
10 solver executions per training update independent of the size of the candidate pool.
11 Across challenging reasoning tasks, our method improves candidate coverage and,
12 in a controlled comparison, achieves performance comparable to multi-draw train-
13 ing with lower training cost.

14 1 Introduction

15 Complex reasoning problems often admit several solution paths, and a single solver execution can
16 fail even when another would succeed. Multi-sample inference reduces this risk by generating sev-
17 eral candidate solutions and selecting or aggregating them, as in repeated sampling, self-consistency,
18 and verifier-guided reasoning [1–3]. With a limited inference budget, the candidate pool succeeds
19 when at least one candidate solves the problem. Repeated sampling can substantially expand prob-
20 lem coverage, although its gains depend on how candidates are generated and selected [4–6]. Recent
21 training methods therefore score groups of outputs jointly or tailor fine-tuning to multi-sample se-
22 lection [7–10]. This line of work treats candidate-pool success as a training target rather than only
23 an inference-time metric.

24 A posterior over solver parameters naturally supplies the multiple candidates used at inference. Each
25 posterior draw instantiates a complete solver that produces one candidate solution, so repeated draws
26 assemble the inference pool from a single learned distribution. We learn this distribution as a reg-
27 ularized posterior, combining task loss with regularization toward a prior [11–13]. The posterior’s
28 probability mass determines which solver variants appear in the pool and how often they are sam-
29 pled, so optimizing it directly shapes the candidate generator used at inference. Variational posterior
30 learning typically optimizes the expected loss of one sampled model. For multi-sample inference,
31 we instead introduce finite- K posterior learning, which trains the regularized posterior to make a
32 finite candidate pool more likely to contain at least one successful solution. Direct optimization of
33 this pool-level objective requires several posterior-sampled solvers in every update, so its cost grows
34 with the target pool size.

35 We reduce this cost with a Fenchel reformulation and a Bregman update that require only one poste-
36 rior draw per training update. The Fenchel reformulation expresses the pool-level objective so that

one sampled solver can update the posterior when its failure is weighted by how likely the other candidates are to fail. The required weight differs across problems and changes as the posterior learns because sampled solvers do not fail equally often on every problem. Estimating it from several fresh draws would cancel the computational saving. We therefore maintain, for each training problem, a running summary of how often its sampled solvers fail. A closed-form Bregman update refreshes this summary after each observed failure, so the current summary guides the posterior update and the new observation informs the next visit. The procedure needs neither an inner optimization loop nor another solver execution. Each update uses one sampled solver regardless of the candidate-pool size.

Across challenging reasoning tasks, our method improves candidate coverage and, in a controlled comparison, achieves performance comparable to direct multi-draw training with lower training cost. Our contributions are:

- We formulate finite- K posterior learning for reasoning, aligning a regularized posterior over solver parameters with the collective success of the finite candidate pool used at inference.
- We derive an exact Fenchel reformulation of the differentiable finite- K objective that enables posterior updates from one sampled solver, guided by a problem-specific summary of posterior failures.
- We derive a closed-form Bregman update for tracking these failure summaries and analyze how sampling noise and posterior drift affect their tracking error and the resulting finite- K objective gap.

2 Related work

2.1 Multi-sample Inference and Test-time Scaling

Multi-sample inference improves performance by generating multiple candidate solutions and aggregating their complementary outcomes. Its benefit was quantified in code generation through Pass@ K , which measures whether at least one sampled program passes the tests [1]. For reasoning tasks, self-consistency aggregates diverse reasoning paths through answer-level voting [2], while outcome and process verifiers rank candidate solutions using learned correctness signals [3, 14]. Structured search methods further extend this principle by expanding and pruning intermediate reasoning branches instead of sampling complete solutions independently [15]. These inference procedures can be viewed as forms of test-time scaling that allocate additional compute to broader candidate exploration. Repeated-sampling studies demonstrate that increasing the number of attempts can substantially expand problem coverage [4], whereas systematic analyses show that the resulting gains depend on the inference algorithm, selector, model size, and allocation of compute across problems [5, 6]. Collectively, this literature establishes the sampling budget K as an important inference-time resource, but generally leaves the candidate-generating distribution unchanged during training.

This train–inference mismatch has motivated inference-aware learning methods that optimize the collective utility of multiple samples. Recent approaches directly optimize K -sample objectives using reinforcement learning [7, 8], tailor fine-tuning to best-of- N selection [9], or modify training to preserve coverage under repeated generation [10]. These methods align a point policy with multi-sample inference, but their updates either operate on groups of output trajectories or exploit autoregressive likelihood structure. Our work instead learns a posterior $q_\phi(\theta)$ over solver parameters while retaining the selector of the underlying reasoning method. We optimize a differentiable surrogate of the finite- K all-fail risk and introduce a Fenchel reformulation with lagged per-problem dual states, such that one posterior draw yields an unbiased gradient for the fixed-dual primal subproblem. This provides a parameter-space formulation of inference-aware learning without executing all K solvers in every primal update.

2.2 Posterior learning for reasoning

Researchers have used posterior distributions over model parameters to represent uncertainty and instantiate multiple neural networks from a single learned distribution. Bissiri et al. generalize

Bayesian updating by combining task losses with KL regularization toward a prior, while loss-calibrated inference adapts posterior approximations to downstream decision utilities [11, 16, 17]. Stochastic variational methods made neural weight distributions trainable through sampled parameters and backpropagation [12, 18], and natural-gradient methods later scaled this approach to modern deep networks [13, 19]. Post-hoc alternatives such as Laplace approximations and SWAG also construct sampleable weight distributions around trained solutions [20, 21]. These methods primarily optimize the expected loss of an individual sampled model or target predictive uncertainty and calibration; their posterior objectives do not account for the collective success of a fixed-size candidate pool.

Recent reasoning methods have begun to use stochastic computation to explore multiple solution paths. GRAM learns distributions over latent reasoning trajectories, whereas PTRM injects noise into recursive latent states without learning a parameter posterior [22, 23]. More closely related approaches move exploration into parameter space: PSN-RLVR perturbs policy weights with an adaptive noise schedule, and 3PO samples one or more policies from an IVON posterior to diversify RLVR rollouts and advantage groups [24, 25]. Although 3PO also learns a distribution over reasoning-model parameters, its generalized variational objective remains linear in the posterior through an expected GRPO loss, and it uses posterior samples to improve training-time exploration. Our method directly trains a solver-parameter posterior for the deployed inference protocol by minimizing the probability that all K posterior-sampled solvers fail, while retaining the prescribed reasoning depth and selector.

2.3 Efficient optimization of multi-sample objectives

Training objectives that evaluate several sampled candidates together are expensive because each update must normally generate the entire candidate set. Direct estimators evaluate at least K samples for a group of size K , even when they reuse subsets of a larger batch to reduce variance [8, 26]. Another line of work reduces this cost by carrying information from earlier updates in a running estimate of the average sample outcome. SCGD introduced this approach for objectives that first average random outcomes and then score the average [27]. NASA later used a single-loop design to track both this average and the model gradient [28]. These methods replace a fresh multi-sample estimate at every update with a state that is refined over time.

For finite training sets, later methods maintain a separate running estimate for each example. SOX updates the estimates associated with the current minibatch and uses their previous values to form the model update, allowing each update to use only a few new samples [29]. ALEXR develops a Fenchel-based method for convex problems, updating a helper variable from one sample batch and the model from an independent batch [30]. For smooth scoring functions, its Bregman update reduces to an ordinary moving average. In our setting, the running state summarizes the recent failure rate of posterior-sampled solvers under fixed- K inference, while the selector remains unchanged. We fix one scalar per problem before the current solver draw, use that draw for the posterior update, and update the scalar afterward. This order gives an unbiased gradient for the update with the scalar held fixed, while our tracking analysis quantifies how sampling noise and posterior changes affect the scalar and the finite- K surrogate gap.

3 Finite- K posterior learning

We formulate the exact all-fail risk under the prescribed inference protocol and obtain a differentiable finite- K surrogate for posterior learning (section 3.1). Fenchel conjugacy introduces a per-problem dual state that makes the surrogate’s posterior-dependent term a one-draw expectation (section 3.2). We update this state from running failure estimates through a Bregman update, and its tracking gap measures the error from using a lagged dual state in the surrogate objective (section 3.3).

3.1 Posterior learning for fixed sampling budgets

Many reasoning problems admit multiple solution paths, so a single solver execution can fail even when another succeeds. Multi-sample inference uses a fixed budget of K executions to increase the chance that the candidate pool contains a successful solution. We make the candidate-generating

139 distribution trainable by learning a posterior $q_\phi(\theta)$ over solver parameters, regularized toward a prior
 140 $p_0(\theta)$. At inference, we draw K parameter samples from this posterior and run each sampled solver
 141 at the prescribed reasoning depth D . The selector associated with the underlying reasoning method
 142 then chooses or aggregates their outputs. We retain that selector and its established training protocol
 143 to isolate posterior learning from selector design. Learning a new selector or co-training the selector
 144 and posterior is a complementary research direction. Our objective targets candidate coverage. The
 145 retained selector determines which candidate the system returns.

146 For problem x , let $F(\theta, x) \in \{0, 1\}$ denote the exact failure indicator for solver θ at depth D , with
 147 $F = 1$ for failure. For independent draws $\theta_{1:K} \stackrel{\text{iid}}{\sim} q_\phi$,

$$\begin{aligned} \Pr(\text{all } K \text{ solvers fail} \mid x) &= \mathbb{E}_{\theta_{1:K}} \left[\prod_{k=1}^K F(\theta_k, x) \right] \\ &= (\mathbb{E}_{\theta \sim q_\phi} [F(\theta, x)])^K. \end{aligned} \quad (1)$$

148 The product is the all-fail indicator. Independence yields the K -th power, so minimizing it maxi-
 149 mizes Pass@ K .

150 This all-fail probability induces the regularized exact target

$$\min_{\phi} \mathbb{E}_x \left[(\mathbb{E}_{\theta \sim q_\phi} [F(\theta, x)])^K \right] + \tau_{\text{KL}} D_{\text{KL}}(q_\phi \| p_0). \quad (2)$$

151 The data term is the expected all-fail probability, and the KL term regularizes the solver posterior
 152 toward p_0 [11, 16].

153 The discrete indicator F does not provide a pathwise gradient through the solver. We therefore
 154 replace it during training with a differentiable failure score $f(\theta, x) \in [0, 1]$ and write $\bar{f}_\phi(x) :=$
 155 $\mathbb{E}_{\theta \sim q_\phi} [f(\theta, x)]$ for its posterior mean. The resulting differentiable training objective is

$$\min_{\phi} \mathbb{E}_x [\bar{f}_\phi(x)^K] + \tau_{\text{KL}} D_{\text{KL}}(q_\phi \| p_0). \quad (3)$$

156 For $K = 1$, its data term reduces to the expected surrogate failure of one posterior draw. When
 157 $f = F$, the optimization problems in eqs. (2) and (3) coincide; otherwise, eq. (3) serves as a smooth
 158 proxy for the exact all-fail risk.

159 3.2 One-draw min-max posterior learning

160 The product of K independent failure scores is an unbiased estimator of $\bar{f}_\phi(x)^K$, but computing it
 161 requires K solver executions per update. A one-draw alternative raises one sampled failure score to
 162 the K -th power. This quantity estimates $\mathbb{E}_{\theta \sim q_\phi} [f(\theta, x)^K]$, whereas the finite- K surrogate contains
 163 $(\mathbb{E}_{\theta \sim q_\phi} [f(\theta, x)])^K$. We therefore seek an exact reformulation of the surrogate in eq. (3) that avoids
 164 K solver executions at every update.

165 The finite- K surrogate has an exact min-max form. This form makes the posterior-dependent update
 166 a one-draw expectation for a fixed dual state.

167 Define $\psi_K(\lambda) = (K - 1)(\lambda/K)^{K/(K-1)}$ on $[0, K]$. For $K \geq 2$ and $r \in [0, 1]$,

$$r^K = \max_{\lambda \in [0, K]} \{\lambda r - \psi_K(\lambda)\}, \quad \lambda^*(r) = K r^{K-1}. \quad (4)$$

168 Applying eq. (4) to $r = \bar{f}_\phi(x)$ gives the equivalent saddle objective

$$\min_{\phi} \max_{\{\lambda_x \in [0, K]\}_x} \left\{ \mathbb{E}_x [\mathbb{E}_{\theta \sim q_\phi} [\lambda_x f(\theta, x)] - \psi_K(\lambda_x)] + \tau_{\text{KL}} D_{\text{KL}}(q_\phi \| p_0) \right\}. \quad (5)$$

169 For a finite training set, eq. (5) requires one scalar dual state λ_x for each training problem.

170 For a fixed current value of λ_x , its posterior-dependent term is an ordinary expectation over $\theta \sim q_\phi$.
 171 Assume that q_ϕ admits a differentiable reparameterization and that differentiation can pass through
 172 the expectation. One posterior draw then gives an unbiased stochastic gradient of the fixed-dual
 173 term. The optimal dual state is $\lambda_x^* = K \bar{f}_\phi(x)^{K-1}$. If $\lambda_x = \lambda_x^*$, the one-draw posterior gradient
 174 equals the finite- K surrogate data gradient. Otherwise, it remains unbiased for the current fixed-
 175 dual objective, whereas it need not equal the gradient of the nonlinear surrogate objective. The KL
 176 regularizer remains part of the posterior update.

3.3 Bregman update for the dual state

The optimal dual state depends on $\bar{f}_\phi(x)$ and changes as the posterior changes. Re-estimating it from multiple posterior draws at every update would remove the one-draw advantage. Instead, we update the dual state from the current sampled failure score $f_t = f(\theta_t, x)$.

Let $D_{\psi_K}(\lambda, \mu) = \psi_K(\lambda) - \psi_K(\mu) - \psi'_K(\mu)(\lambda - \mu)$ denote the Bregman divergence generated by ψ_K . We apply the Bregman-proximal ascent step

$$\lambda_{x,t+1} = \arg \max_{\lambda \in [0, K]} \{ \lambda f_t - \psi_K(\lambda) - \eta^{-1} D_{\psi_K}(\lambda, \lambda_{x,t}) \}, \quad \eta > 0. \quad (6)$$

Since $\psi'_K(\lambda) = (\lambda/K)^{1/(K-1)}$, this update has a closed form. With $\beta = 1/(1 + \eta)$ and $\hat{f}_{x,t-1} = \psi'_K(\lambda_{x,t})$,

$$\begin{aligned} \hat{f}_{x,t} &= \beta \hat{f}_{x,t-1} + (1 - \beta) f_t, \\ \lambda_{x,t+1} &= K \hat{f}_{x,t}^{K-1}. \end{aligned} \quad (7)$$

The closed form reduces dual ascent to a scalar exponential moving average (EMA). It requires neither an inner optimization loop nor an additional solver execution. A larger β suppresses sampling noise but increases lag when the posterior failure rate changes.

At update t , we compute $\lambda_{x,t} = K \hat{f}_{x,t-1}^{K-1}$ from the running estimate available before drawing θ_t . We hold this state fixed during the posterior update. Afterward, we incorporate f_t into $\hat{f}_{x,t}$ and use the result at the next update. This order fixes the dual state before the current posterior sample, which gives the conditional one-draw gradient property in section 3.2.

Algorithm 1 One-draw finite- K posterior learning

Require: Posterior q_ϕ , prior p_0 , pool size K , coefficient β , and failure estimates $\{\hat{f}_x\}$
1: **for** each training update t **do**
2: Set and detach $\lambda_{x,t} \leftarrow K \hat{f}_{x,t-1}^{K-1}$ for each minibatch problem x
3: Draw $\theta_t \sim q_\phi$ once and compute $f(\theta_t, x)$
4: Update ϕ with $\lambda_{x,t} f(\theta_t, x)$ and the KL regularizer while holding $\lambda_{x,t}$ fixed
5: Update $\hat{f}_{x,t} \leftarrow \beta \hat{f}_{x,t-1} + (1 - \beta) f(\theta_t, x)$
6: **end for**

Direct finite- K training uses K solver executions per update. The exact min-max reformulation makes the posterior update a one-draw expectation once the current dual state is held fixed. The Bregman step gives a closed-form EMA update for that state without additional solver executions. The practical update follows the finite- K data gradient to the extent that $\hat{f}_{x,t-1}$ tracks $\bar{f}_\phi(x)$.

4 Experiments

4.1 Experimental setup

Datasets. We use Sudoku-Extreme, a set of difficult 9×9 Sudoku completion problems, and Maze-Hard, to evaluate optimal path finding in 30×30 grid mazes. For ARC-AGI-2, the model infers an abstract transformation from a small set of input-output grid pairs and applies it to held-out inputs [31].

Baselines. We compare FB with deterministic reasoning models (HRM, TRM, and FPRM), sampling-based reasoning methods (GRAM, PTRM, W-PTRM, and W-FPRM), and posterior optimization methods (IVON and FB). Detailed method descriptions, initialization, and fine-tuning protocols are provided in appendix B.

4.2 Main results: Finite- K posterior learning improves reasoning

Table 1 shows that FB obtains 99.53% selected accuracy and 99.55% Pass@10 on Sudoku-Extreme. These values exceed W-PTRM, the strongest listed comparator for both metrics, by 0.70 and 0.71

Table 1: **FB achieves the highest selected accuracy and Pass@10 on Sudoku-Extreme and Maze-Hard.** Selected accuracy uses each method’s native selector, while Pass@10 measures exact solution coverage among ten candidates. Bold and underlined values mark the best and second-best completed results. [†] All HRM values are paper-reported.

Method	Sudoku-Extreme		Maze-Hard		ARC-AGI-2		
	Acc.	Pass@10	Acc.	Pass@10	Pass@1	Pass@2	Pass@100
HRM [†]	55.00	–	74.50	–	–	5.00	–
FPRM	94.20	–	86.90	–	–	–	–
GRAM	98.48	98.49	87.30	87.30	–	–	–
TRM	87.28	–	87.00	–	2.92	4.58	10.83
PTRM	97.36	97.38	<u>87.70</u>	88.40	2.92	3.75	10.83
W-PTRM	<u>98.83</u>	<u>98.84</u>	87.50	<u>89.80</u>	2.92	3.75	11.25
FB (Ours)	99.53	99.55	88.00	91.00	pending	pending	pending

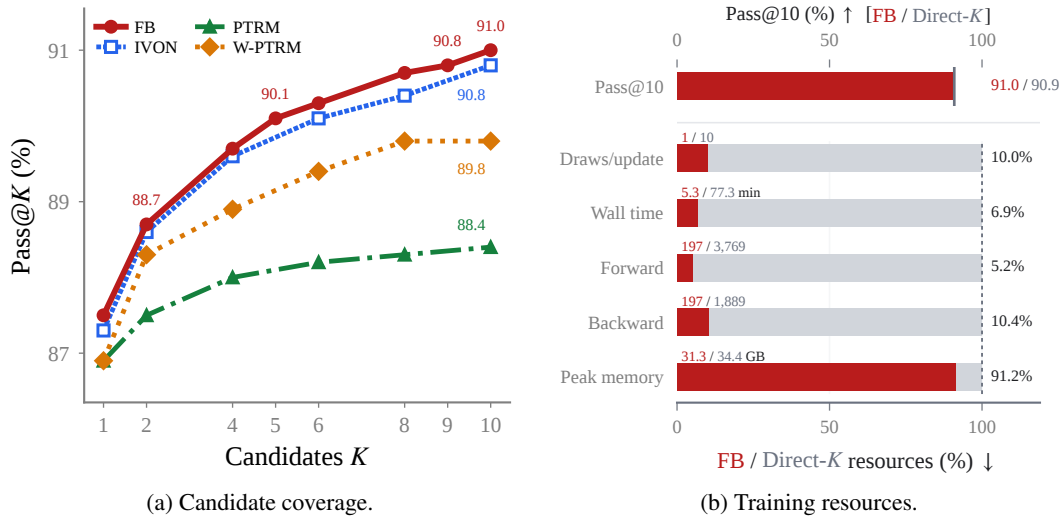


Figure 1: **FB improves inference and training efficiency on Maze-Hard.** (a) FB reaches each baseline’s Pass@10 with fewer candidates using the same TRM backbone, 1,000 test problems, and reasoning depth 16. (b) At 188 updates, one-draw FB maintains comparable Pass@10 to Direct- $K=10$ while reducing training resources. Direct- K uses seed 0; FB reports the mean over seeds 0–2.

percentage points. On Maze-Hard, FB obtains 88.00% selected accuracy and 91.00% Pass@10. The corresponding gains are 0.30 points over PTRM in selected accuracy and 1.20 points over W-PTRM in Pass@10.

The difference between Pass@10 and selected accuracy is 0.02 points on Sudoku-Extreme and 3.00 points on Maze-Hard. Thus, on Maze-Hard, the candidate pool contains an exact solution for more puzzles than the native selector solves. [RESULT PENDING: We defer the ARC-AGI-2 comparison until the FB evaluation is complete.]

Candidate coverage under fixed inference budgets. Pass@ K measures how candidate coverage changes as the inference budget grows. We compare FB, ordinary IVON, PTRM, and W-PTRM on the same TRM backbone and Maze-Hard test set at reasoning depth 16. Figure 1a reports Pass@ K at $K \in \{1, 2, 4, 6, 8, 10\}$.

Using all saved FB prefixes to locate the first crossing, FB reaches PTRM’s Pass@10 of 88.4% with two candidates, reducing solver executions by 80%, and reaches W-PTRM’s 89.8% with five candidates, reducing them by 50%. FB also reaches IVON’s Pass@10 of 90.8% with nine candidates,

223 reducing solver executions by 10%; across the displayed budgets, FB exceeds IVON by 0.1–0.3
224 percentage points.

225 4.3 Fidelity of one-draw finite- K optimization

226 We test whether one-draw FB follows the original finite- K objective during training.
227 [PLACEHOLDER: Compare direct K -draw optimization, the naive one-sample surrogate, and FB
228 from matched initialization and training data. Add a three-panel figure reporting (a) the true finite- K
229 objective estimated with multiple posterior draws, (b) the saddle-objective gap, and (c) dual tracking
230 error over training.]

231 4.4 Computational advantage of one-draw training

232 Direct finite- K optimization requires K solver executions per update, whereas FB uses one. At
233 the matched endpoint of 188 updates, Figure 1b compares Direct- $K=10$ seed 0 with the mean of
234 FB seeds 0–2. FB attains 91.0% Pass@10 versus 90.9% for Direct- K , while using 10.0% as many
235 posterior draws per update, 6.9% of the wall-clock time, 5.2% of the forward passes, and 10.4% of
236 the backward passes. Peak memory is reduced more modestly to 91.2%. This endpoint comparison
237 supports comparable performance with substantially less training compute; it does not establish a
238 cost-to-target result.

239 5 Limitations

240 6 Conclusion

References

- [1] Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, Alex Ray, Raul Puri, Gretchen Krueger, Michael Petrov, Heidy Khlaaf, Girish Sastry, Pamela Mishkin, Brooke Chan, Scott Gray, Nick Ryder, Mikhail Pavlov, Alethea Power, Lukasz Kaiser, Mohammad Bavarian, Clemens Winter, Philippe Tillet, Felipe Petroski Such, Dave Cummings, Matthias Plappert, Fotios Chantzis, Elizabeth Barnes, Ariel Herbert-Voss, William Hebgen Guss, Alex Nichol, Alex Paino, Nikolas Tezak, Jie Tang, Igor Babuschkin, Suchir Balaji, Shantanu Jain, William Saunders, Christopher Hesse, Andrew N. Carr, Jan Leike, Josh Achiam, Vedant Misra, Evan Morikawa, Alec Radford, Matthew Knight, Miles Brundage, Mira Murati, Katie Mayer, Peter Welinder, Bob McGrew, Dario Amodei, Sam McCandlish, Ilya Sutskever, and Wojciech Zaremba. Evaluating large language models trained on code, 2021. URL <https://arxiv.org/abs/2107.03374>. arXiv:2107.03374.
- [2] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V. Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=1PL1NIMMrw>.
- [3] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Rei Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems, 2021. URL <https://arxiv.org/abs/2110.14168>. arXiv:2110.14168.
- [4] Bradley Brown, Jordan Juravsky, Ryan Ehrlich, Ronald Clark, Quoc V. Le, Christopher Ré, and Azalia Mirhoseini. Large language monkeys: Scaling inference compute with repeated sampling, 2024. URL <https://arxiv.org/abs/2407.21787>. arXiv:2407.21787.
- [5] Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. Scaling LLM test-time compute optimally can be more effective than scaling model parameters. In *International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=4FWAwZtd2n>.
- [6] Yangzhen Wu, Zhiqing Sun, Shanda Li, Sean Welleck, and Yiming Yang. Inference scaling laws: An empirical analysis of compute-optimal inference for LLM problem-solving. In *International Conference on Learning Representations*, pages 55824–55845, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/file/8c3caae2f725c8e2a55ecd600563d172-Paper-Conference.pdf.
- [7] Yunhao Tang, Kunhao Zheng, Gabriel Synnaeve, and Remi Munos. Optimizing language models for inference time objectives using reinforcement learning. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 59066–59085, 2025. URL <https://proceedings.mlr.press/v267/tang25o.html>.
- [8] Christian Walder and Deep Karkhanis. Pass@K policy optimization: Solving harder reinforcement learning problems, 2025. URL <https://arxiv.org/abs/2505.15201>. arXiv:2505.15201.
- [9] Yinlam Chow, Guy Tennenholtz, Izzeddin Gur, Vincent Zhuang, Bo Dai, Aviral Kumar, Rishabh Agarwal, Sridhar Thiagarajan, Craig Boutilier, and Aleksandra Faust. Inference-aware fine-tuning for best-of-N sampling in large language models. In *International Conference on Learning Representations*, pages 78936–78959, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/file/c40bed606c51c8e827c1ba75aa2da054-Paper-Conference.pdf.
- [10] Feng Chen, Allan Raventós, Nan Cheng, Surya Ganguli, and Shaul Druckmann. Rethinking fine-tuning when scaling test-time compute: Limiting confidence improves mathematical reasoning. In *Advances in Neural Information Processing Systems*, volume 38, pages 158785–158818, 2025. doi: 10.52202/085713-5304. URL https://proceedings.neurips.cc/paper_files/paper/2025/file/e8f4eae0a41cab67fdead3aa6b77f083-Paper-Conference.pdf.

- [11] Pier Giovanni Bissiri, Chris C. Holmes, and Stephen G. Walker. A general framework for updating belief distributions. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 78(5):1103–1130, 2016. doi: 10.1111/rssb.12158.
- [12] Charles Blundell, Julien Cornebise, Koray Kavukcuoglu, and Daan Wierstra. Weight uncertainty in neural network. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pages 1613–1622, 2015. URL <https://proceedings.mlr.press/v37/blundell115.html>.
- [13] Yuesong Shen, Nico Daheim, Bai Cong, Peter Nickl, Gian Maria Marconi, Bazan Clement Emile Marcel Raoul, Rio Yokota, Iryna Gurevych, Daniel Cremers, Mohammad Emtiyaz Khan, and Thomas Möllenhoff. Variational learning is effective for large deep networks. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 44665–44686, 2024. URL <https://proceedings.mlr.press/v235/shen24b.html>.
- [14] Hunter Lightman, Vineet Kosaraju, Yuri Burda, Harrison Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s verify step by step. In *International Conference on Learning Representations*, pages 39578–39601, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/file/aca97732e30bcf1303bc22ac3924fd16-Paper-Conference.pdf.
- [15] Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Tom Griffiths, Yuan Cao, and Karthik Narasimhan. Tree of thoughts: Deliberate problem solving with large language models. In *Advances in Neural Information Processing Systems*, volume 36, pages 11809–11822, 2023. doi: 10.52202/075280-0517. URL https://proceedings.neurips.cc/paper_files/paper/2023/file/271db9922b8d1f4dd7aaef84ed5ac703-Paper-Conference.pdf.
- [16] Simon Lacoste-Julien, Ferenc Huszár, and Zoubin Ghahramani. Approximate inference for the loss-calibrated bayesian. In *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, volume 15 of *Proceedings of Machine Learning Research*, pages 416–424, 2011. URL <https://proceedings.mlr.press/v15/lacoste-julien11a.html>.
- [17] Adam D. Cobb, Stephen J. Roberts, and Yarin Gal. Loss-calibrated approximate inference in bayesian neural networks, 2018. URL <https://arxiv.org/abs/1805.03901>. arXiv:1805.03901.
- [18] Alex Graves. Practical variational inference for neural networks. In *Advances in Neural Information Processing Systems*, volume 24, pages 2348–2356, 2011. URL https://papers.nips.cc/paper_files/paper/2011/hash/7eb3c8be3d411e8ebfab08eba5f49632-Abstract.html.
- [19] Kazuki Osawa, Siddharth Swaroop, Mohammad Emtiyaz Khan, Anirudh Jain, Runa Eschenhagen, Richard E. Turner, and Rio Yokota. Practical deep learning with bayesian principles. In *Advances in Neural Information Processing Systems*, volume 32, 2019. URL https://proceedings.neurips.cc/paper_files/paper/2019/hash/b53477c2821c1bf0da5d40e57b870d35-Abstract.html.
- [20] Erik Daxberger, Agustinus Kristiadi, Alexander Immer, Runa Eschenhagen, Matthias Bauer, and Philipp Hennig. Laplace redux: Effortless bayesian deep learning. In *Advances in Neural Information Processing Systems*, volume 34, 2021. URL https://proceedings.neurips.cc/paper_files/paper/2021/hash/a7c9585703d275249f30a088cebba0ad-Abstract.html.
- [21] Wesley J. Maddox, Pavel Izmailov, Timur Garipov, Dmitry P. Vetrov, and Andrew Gordon Wilson. A simple baseline for bayesian uncertainty in deep learning. In *Advances in Neural Information Processing Systems*, volume 32, 2019. URL <https://papers.nips.cc/paper/2019/hash/118921efba23fc329e6560b27861f0c2-Abstract.html>.
- [22] Junyeob Baek, Mingyu Jo, Minsu Kim, Mengye Ren, Yoshua Bengio, and Sungjin Ahn. Generative recursive reasoning, 2026. URL <https://arxiv.org/abs/2605.19376>. arXiv:2605.19376.

- [23] Amin Sghaier, Ali Parviz, and Alexia Jolicoeur-Martineau. Probabilistic tiny recursive model, 2026. URL <https://arxiv.org/abs/2605.19943>. arXiv:2605.19943.
- [24] Bizhe Bai, Xinyue Wang, Peng Ye, and Tao Chen. Learning to explore with parameter-space noise: A deep dive into parameter-space noise for reinforcement learning with verifiable rewards, 2026. URL <https://arxiv.org/abs/2602.02555>. arXiv:2602.02555, version 3.
- [25] Vatsal Venkatkrishna, Nico Daheim, and Iryna Gurevych. Parameter exploration for RLVR via variational learning, 2026. URL <https://arxiv.org/abs/2608.09805>. arXiv:2608.09805, version 1.
- [26] Wassily Hoeffding. A class of statistics with asymptotically normal distribution. *The Annals of Mathematical Statistics*, 19(3):293–325, 1948. doi: 10.1214/aoms/1177730196. URL <https://doi.org/10.1214/aoms/1177730196>.
- [27] Mengdi Wang, Ethan X. Fang, and Han Liu. Stochastic compositional gradient descent: Algorithms for minimizing compositions of expected-value functions. *Mathematical Programming*, 161(1–2):419–449, 2017. doi: 10.1007/s10107-016-1017-3. URL <https://doi.org/10.1007/s10107-016-1017-3>.
- [28] Saeed Ghadimi, Andrzej Ruszczyński, and Mengdi Wang. A single timescale stochastic approximation method for nested stochastic optimization. *SIAM Journal on Optimization*, 30(1):960–979, 2020. doi: 10.1137/18M1230542. URL <https://doi.org/10.1137/18M1230542>.
- [29] Bokun Wang and Tianbao Yang. Finite-sum coupled compositional stochastic optimization: Theory and applications. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 23292–23317, 2022. URL <https://proceedings.mlr.press/v162/wang22ak.html>.
- [30] Bokun Wang and Tianbao Yang. A near-optimal single-loop stochastic algorithm for convex finite-sum coupled compositional optimization. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 65091–65121, 2025. URL <https://proceedings.mlr.press/v267/wang25dw.html>.
- [31] François Chollet, Mike Knoop, Gregory Kamradt, Bryan Landers, and Henry Pinkard. ARC-AGI-2: A new challenge for frontier AI reasoning systems, 2025. URL <https://arxiv.org/abs/2505.11831>. arXiv:2505.11831.
- [32] Guan Wang, Jin Li, Yuhao Sun, Xing Chen, Changling Liu, Yue Wu, Meng Lu, Sen Song, and Yasin Abbasi Yadkori. Hierarchical reasoning model, 2025. URL <https://arxiv.org/abs/2506.21734>. arXiv:2506.21734.
- [33] Alexia Jolicoeur-Martineau. Less is more: Recursive reasoning with tiny networks, 2025. URL <https://arxiv.org/abs/2510.04871>. arXiv:2510.04871.
- [34] Sajad Movahedi, Vera Milovanović, Shlomo Libo Feigin, Alexander Theus, Thomas Hofmann, Valentina Boeva, T. Konstantin Rusch, and Antonio Orvieto. Fixed-point reasoners: Stable and adaptive deep looped transformers, 2026. URL <https://arxiv.org/abs/2606.18206>. arXiv:2606.18206.

386 A Finite- K and dual-state derivations

387 A.1 Binary target and direct Monte Carlo estimation

388 Let $p_\phi(x) = \mathbb{E}_{\theta \sim q_\phi}[F(\theta, x)]$ denote the failure probability of one posterior-sampled solver. For
 389 independent draws $\theta_{1:K} \sim q_\phi$,

$$\Pr(\text{all } K \text{ draws fail} \mid x) = \mathbb{E} \left[\prod_{k=1}^K F(\theta_k, x) \right] = p_\phi(x)^K. \quad (8)$$

390 Hence, the probability of at least one success is $1 - p_\phi(x)^K$, which is the corresponding Pass@ K
 391 probability. For binary indicators, the product equals the indicator that all K draws fail. The product
 392 therefore represents the deployment event rather than an additional modeling assumption.

393 Replacing F with the differentiable score f preserves the product across independent draws. Inde-
 394 pendence gives

$$\mathbb{E} \left[\prod_{k=1}^K f(\theta_k, x) \right] = \prod_{k=1}^K \mathbb{E}[f(\theta_k, x)] = \bar{f}_\phi(x)^K. \quad (9)$$

395 Thus, the K -draw product is an unbiased estimator of the finite- K data term. Reusing one draw
 396 estimates $\mathbb{E}[f(\theta, x)^K]$. For $K > 1$, Jensen’s inequality gives $\mathbb{E}[f^K] \geq \mathbb{E}[f]^K$, with equality only in
 397 special cases. In the binary limit, $F^K = F$, so the reused-draw objective reduces to the one-draw
 398 failure probability.

399 A.2 Derivation of the Fenchel representation

400 For $K \geq 2$, consider $h_r(\lambda) = \lambda r - \psi_K(\lambda)$ on $[0, K]$. Its derivative on $(0, K)$ is

$$h'_r(\lambda) = r - \left(\frac{\lambda}{K} \right)^{1/(K-1)}. \quad (10)$$

401 The unique stationary point is $\lambda^*(r) = Kr^{K-1}$, which lies in $[0, K]$ for $r \in [0, 1]$. Substitution
 402 gives

$$h_r(\lambda^*(r)) = Kr^{K-1}r - (K-1)r^K = r^K. \quad (11)$$

403 Because ψ_K is strictly convex, h_r is strictly concave and this stationary point is the unique maxi-
 404 mizer for $r \in (0, 1)$. The boundary cases give $\lambda^*(0) = 0$ and $\lambda^*(1) = K$. The case $K = 1$ needs
 405 no Fenchel step because the objective is linear in r .

406 A.3 Fixed-dual one-draw gradient

407 Assume that q_ϕ admits a differentiable reparameterization $\theta = T_\phi(\epsilon)$ whose base distribution does
 408 not depend on ϕ . Also assume that an integrable dominating function permits differentiation under
 409 the expectation. For a fixed detached dual state λ_x ,

$$\begin{aligned} \nabla_\phi \mathbb{E}_{q_\phi}[\lambda_x f(\theta, x)] &= \nabla_\phi \mathbb{E}_\epsilon[\lambda_x f(T_\phi(\epsilon), x)] \\ &= \mathbb{E}_\epsilon[\nabla_\phi \{\lambda_x f(T_\phi(\epsilon), x)\}]. \end{aligned} \quad (12)$$

410 One draw of ϵ , or equivalently one $\theta \sim q_\phi$, therefore gives an unbiased stochastic gradient of the
 411 fixed-dual data term. Moreover,

$$\nabla_\phi \bar{f}_\phi(x)^K = K \bar{f}_\phi(x)^{K-1} \nabla_\phi \bar{f}_\phi(x) = \lambda^*(\bar{f}_\phi(x)) \nabla_\phi \bar{f}_\phi(x). \quad (13)$$

412 The fixed-dual gradient coincides with the finite- K data gradient when $\lambda_x = \lambda^*(\bar{f}_\phi(x))$. For a
 413 lagged dual state, unbiasedness holds conditionally for the fixed-dual objective.

414 A.4 Dual coordinate and Bregman update

415 **Lemma 1** (Dual coordinate). *For integer $K \geq 2$ and $\lambda \in [0, K]$, the derivative $\psi'_K(\lambda) =$
 416 $(\lambda/K)^{1/(K-1)}$ is a strictly increasing bijection from $[0, K]$ to $[0, 1]$. Its inverse is $(\psi'_K)^{-1}(u) =$
 417 Ku^{K-1} for $u \in [0, 1]$, and $\psi'_K(\lambda^*(r)) = r$.*

418 *Proof.* Direct differentiation gives the stated derivative, including its continuous extension at zero.
 419 The derivative increases from zero to one on $[0, K]$. Solving $u = (\lambda/K)^{1/(K-1)}$ gives the inverse,
 420 and substituting $\lambda^*(r) = Kr^{K-1}$ gives the final identity. $\square$

421 **Proposition 1** (Bregman-proximal dual update). *Let $K \geq 2$, $\eta > 0$, $\lambda_{x,t} \in [0, K]$, and $f_t \in [0, 1]$.
 422 If $\hat{f}_{x,t-1} = \psi'_K(\lambda_{x,t})$ and $\beta = 1/(1 + \eta)$, then the unique maximizer in eq. (6) is given by eq. (7).*

423 *Proof.* Expanding the Bregman divergence in eq. (6) and dropping terms independent of λ gives

$$Q_t(\lambda) = \left(f_t + \frac{\hat{f}_{x,t-1}}{\eta} \right) \lambda - \left(1 + \frac{1}{\eta} \right) \psi_K(\lambda) + \text{const.} \quad (14)$$

424 Strict convexity of ψ_K makes Q_t strictly concave. The first-order condition gives

$$\psi'_K(\lambda_{x,t+1}) = \frac{\hat{f}_{x,t-1} + \eta f_t}{1 + \eta} = \beta \hat{f}_{x,t-1} + (1 - \beta) f_t. \quad (15)$$

425 The right-hand side lies in $[0, 1]$, so lemma 1 covers the interior and both boundary cases. Applying
 426 $(\psi'_K)^{-1}$ yields $\lambda_{x,t+1} = K \hat{f}_{x,t}^{K-1}$. $\square$

427 A.5 Indexing and the lagged dual state

428 The implementation follows the order

$$\hat{f}_{x,t-1} \longrightarrow \lambda_{x,t} = K \hat{f}_{x,t-1}^{K-1} \longrightarrow \theta_t \sim q_{\phi_t} \longrightarrow f_{x,t} \longrightarrow \hat{f}_{x,t}. \quad (16)$$

429 Let $\mathcal{H}_{t-1}$ denote the training history before drawing θ_t . The state $\lambda_{x,t}$ is measurable with respect
 430 to $\mathcal{H}_{t-1}$ and is fixed before the current posterior sample. The one-draw gradient result therefore
 431 applies conditionally on this history. Detaching a dual state formed from the current sample blocks
 432 its autograd path but does not make it statistically independent of that sample. Using the lagged state
 433 avoids this dependence.

434 A.6 Stationary stochastic tracking

435 **Proposition 2** (Stationary stochastic tracking). *Fix $K \geq 2$ and $\beta \in (0, 1)$. Let $f_t \in [0, 1]$ satisfy
 436 $\mathbb{E}[f_t \mid \mathcal{H}_{t-1}] = r$ and $\mathbb{E}[(f_t - r)^2 \mid \mathcal{H}_{t-1}] \leq \sigma^2$ for fixed $r \in [0, 1]$. For $\hat{f}_t = \beta \hat{f}_{t-1} + (1 - \beta) f_t$,
 437 $\lambda_{t+1} = K \hat{f}_t^{K-1}$, and $\lambda^* = Kr^{K-1}$,*

$$\mathbb{E}[\hat{f}_t] - r = \beta^t (\hat{f}_0 - r), \quad (17)$$

$$\mathbb{E}[(\hat{f}_t - r)^2] \leq \beta^{2t} (\hat{f}_0 - r)^2 + \sigma^2 \frac{1 - \beta}{1 + \beta} (1 - \beta^{2t}), \quad (18)$$

$$\mathbb{E}[(\lambda_{t+1} - \lambda^*)^2] \leq K^2 (K - 1)^2 \mathbb{E}[(\hat{f}_t - r)^2]. \quad (19)$$

438 *The Fenchel inner-max gap $\Delta_t = r^K - [\lambda_{t+1} r - \psi_K(\lambda_{t+1})]$ satisfies*

$$0 \leq \mathbb{E}[\Delta_t] \leq \frac{K(K-1)}{2} \mathbb{E}[(\hat{f}_t - r)^2]. \quad (20)$$

439 *Proof.* Set $e_t = \hat{f}_t - r$ and $\xi_t = f_t - r$. Then $e_t = \beta e_{t-1} + (1 - \beta) \xi_t$ and $\mathbb{E}[\xi_t \mid \mathcal{H}_{t-1}] = 0$.
 440 Taking expectations gives eq. (17). Squaring, conditioning, and unrolling the resulting recursion
 441 gives eq. (18). For $g(u) = Ku^{K-1}$ on $[0, 1]$, the bound $|g'(u)| \leq K(K-1)$ and the mean-value
 442 theorem give eq. (19). Finally, $\psi'_K(\lambda^*) = r$ implies $\Delta_t = D_{\psi_K}(\lambda_{t+1}, \lambda^*)$. Writing $u = \hat{f}_t$ and
 443 integrating the derivative of $r^K - Kr u^{K-1} + (K-1)u^K$ between r and u gives eq. (20). $\square$

444 A.7 Non-stationary stochastic tracking

445 **Proposition 3** (Non-stationary stochastic tracking). *Fix $K \geq 2$ and $\beta \in (0, 1)$. Let $r_t = \mathbb{E}[f_t \mid$
446 $\mathcal{H}_{t-1}]$, $\xi_t = f_t - r_t$, and $\mathbb{E}[\xi_t^2 \mid \mathcal{H}_{t-1}] \leq \sigma_t^2$. Define $e_t = \hat{f}_{t-1} - r_t$, $\lambda_t = K\hat{f}_{t-1}^{K-1}$, and
447 $\lambda_t^* = Kr_t^{K-1}$. Then*

$$e_t = \beta^{t-1}e_1 + (1 - \beta) \sum_{j=1}^{t-1} \beta^{t-1-j} \xi_j - \sum_{j=1}^{t-1} \beta^{t-1-j} (r_{j+1} - r_j). \quad (21)$$

448 Consequently,

$$\begin{aligned} \|e_t\|_2 &\leq \beta^{t-1} \|e_1\|_2 + \sum_{j=1}^{t-1} \beta^{t-1-j} \|r_{j+1} - r_j\|_2 \\ &\quad + (1 - \beta) \left(\sum_{j=1}^{t-1} \beta^{2(t-1-j)} \sigma_j^2 \right)^{1/2}. \end{aligned} \quad (22)$$

449 If $\|r_{t+1} - r_t\|_2 \leq \delta$ and $\sigma_t \leq \sigma$, then

$$\limsup_{t \rightarrow \infty} \|e_t\|_2 \leq \frac{\delta}{1 - \beta} + \sigma \sqrt{\frac{1 - \beta}{1 + \beta}}. \quad (23)$$

450 With $\rho_t = \max\{\hat{f}_{t-1}, r_t\}$, the dual error and instantaneous Fenchel gap $\Gamma_t = r_t^K - [\lambda_t r_t - \psi_K(\lambda_t)]$
451 satisfy

$$|\lambda_t - \lambda_t^*| \leq K(K - 1) \rho_t^{K-2} |e_t|, \quad (24)$$

$$0 \leq \Gamma_t \leq \frac{K(K - 1)}{2} \rho_t^{K-2} e_t^2. \quad (25)$$

452 *Proof.* The EMA recursion gives $e_{t+1} = \beta e_t + (1 - \beta) \xi_t - (r_{t+1} - r_t)$. Unrolling it yields eq. (21).
453 The martingale-difference property removes cross terms in the squared norm of the noise sum. The
454 triangle inequality in L_2 then gives eq. (22). Bounding the drift and noise terms by geometric
455 series gives eq. (23). Factoring the difference between $\hat{f}_{t-1}^{K-1}$ and r_t^{K-1} gives eq. (24). The identity
456 $\Gamma_t = D_{\psi_K}(\lambda_t, \lambda_t^*)$ and the same integral argument as in proposition 2 give eq. (25). $\square$

457 The bounds separate sampling noise from drift in the posterior failure rate. Increasing β reduces the
458 noise term but increases the lag term $\delta/(1 - \beta)$. These results guarantee tracking of the dual state,
459 not convergence of the full neural-network optimization.

460 B Experimental details

461 B.1 Datasets and evaluation protocols

462 **[PLACEHOLDER:** Report dataset splits, preprocessing, reasoning depth, selectors, the distinc-
463 tion between K_{train} and K_{eval} , evaluation seeds, and confidence-interval construction for Sudoku-
464 Extreme, Maze-Hard, and ARC-AGI-2.]

465 **Maze-Hard candidate-coverage protocol.** We evaluate FB, ordinary IVON, PTRM, and W-
466 PTRM on the same 1,000-problem test split with the TRM backbone, reasoning depth 16, and
467 candidate budgets up to ten. All four methods use ordered prefixes from saved $K_{\text{eval}} = 10$ banks
468 under seed-0 reset sampling. For IVON, we evaluate the EMA model state stored in the seed-0
469 checkpoint.

470 B.2 Baseline details

471 **Deterministic reasoning models.** **HRM** couples recurrent modules that operate at different time
472 scales [32]. **TRM** replaces the hierarchy with one small recursive network [33]. **FPRM** uses fixed-
473 point convergence to halt a looped Transformer [34].

Table 2: **FB obtains the highest selected accuracy and Pass@10 with both backbones on Maze-Hard.** Results are percentages on the complete 1,000-puzzle test set. Bold and underlined values mark the best and second-best results within each backbone.

Method	Acc. $\uparrow$	Pass@10 $\uparrow$
<i>TRM backbone</i>		
Deterministic	87.00	87.00
PTRM	<u>87.70</u>	88.40
W-PTRM	87.50	<u>89.80</u>
FB (Ours)	88.00	91.00
<i>FPRM backbone</i>		
Deterministic	<u>86.90</u>	—
W-FPRM	<u>86.90</u>	88.70
IVON	84.40	<u>94.00</u>
FB (Ours)	87.70	94.80

Sampling-based reasoning methods. GRAM models reasoning as stochastic latent trajectories and samples several trajectories in parallel [22]. PTRM keeps the model parameters fixed, injects Gaussian noise into the latent state at each deep-recursion step, and selects among the resulting trajectories [23]. W-PTRM and W-FPRM are non-learned baselines that sample each candidate’s parameters from a fixed Gaussian centered at the released TRM or FPRM checkpoint. They add no latent noise.

Posterior optimization methods. IVON (Improved Variational Online Newton) maintains a sampleable diagonal Gaussian through an online variational update and optimizes the single-draw posterior objective [13]. FB (Ours) uses the same IVON posterior parameterization and initialization, but optimizes the finite- K objective with Fenchel–Bregman updates. For learned posterior methods, we initialize the posterior mean from a released task checkpoint and fine-tune its mean and diagonal precision on the task training split. Fine-tuning adapts the solver distribution to the chosen objective, while W-PTRM and W-FPRM isolate sampling without this adaptation.

B.3 Training and compute accounting

[PLACEHOLDER: Report optimization hyperparameters, posterior and dual-state initialization, training-update budgets, hardware, batch sizes, mixed-precision settings, solver-call accounting, wall-clock measurement, and peak-memory measurement.]

Ordinary-IVON baseline. Ordinary IVON uses the shared released-checkpoint initialization, task data, update budget, and fixed $K = 10$ test protocol described above.

C Additional posterior analyses

C.1 Posterior learning across recursive backbones

Table 2 compares posterior methods with TRM and FPRM backbones on the complete Maze-Hard test set. Every stochastic method uses ten candidates ($K = 10$) and its native selector.

FB improves both metrics under the two tested backbones. With TRM, it raises selected accuracy by 0.30 percentage points over PTRM and Pass@10 by 1.20 points over W-PTRM. With FPRM, FB improves selected accuracy from 84.40% to 87.70% and Pass@10 from 94.00% to 94.80% over IVON. The paired 95% bootstrap intervals are [1.60, 5.10] for selected accuracy and [0.00, 1.60] for Pass@10. The two FPRM posterior rows use the same initialization, training data, update budget, and evaluation noise.

C.2 Posterior construction under a fixed backbone

Table 3 fixes the FPRM backbone and Maze-Hard protocol while varying how each method constructs its parameter distribution. Diagonal Laplace fits a local Gaussian from curvature at a point estimate [20], while SWAG estimates a Gaussian from an SGD trajectory [21].

Table 3: The comparison fixes FPRM and Maze-Hard while varying the posterior construction.

Method	Acc. $\uparrow$	Pass@10 $\uparrow$
FPRM	86.90	–
W-FPRM	86.90	88.70
Diagonal Laplace	pending	pending
SWAG	pending	pending
IVON	84.40	94.00
FB (Ours)	87.70	94.80

Among posterior results, FB improves selected accuracy by 3.30 percentage points and Pass@10 by 0.80 points over IVON. [RESULT PENDING: Compare FB with Diagonal Laplace and SWAG after their matched Maze-Hard results and fit times are available.]

Candidate diversity. [PLACEHOLDER: Report the unique-solution ratio and candidate disagreement for each posterior method. Pair each diversity statistic with Pass@ K so that output variation is not interpreted as useful coverage by itself.]

D Additional objective and resource analyses

D.1 Extended objective and tracking diagnostics

[PLACEHOLDER: Report the naive one-sample surrogate, detailed saddle-objective gap, per-problem and aggregate dual tracking errors, and sensitivity of the tracking gap over training.]

D.2 Extended inference-budget results

[PLACEHOLDER: Report all saved prefixes for $K = 1, \dots, 10$, per-seed results, confidence intervals, and additional sampling baselines that do not fit in the main figure.]

E Ablation studies

EMA coefficient. [PLACEHOLDER: Evaluate $\beta \in \{0, 0.9, 0.99, 0.999\}$ using dual tracking error, the final finite- K objective, optimization stability, and Pass@ K . Relate smaller β to responsive but noisy estimates and larger β to smoother but slower tracking.]

Other optimization choices. [PLACEHOLDER: Report ablations for the KL coefficient τ_{KL} , dual-state initialization, and K_{train} . Include the final finite- K objective, tracking error, Pass@ K , and selected accuracy where applicable.]